Improving nutrient Mx in rice/wheat rotations to reduce N2O emissions

This example script uses the data to identify possible district that have high rice/wheat rotation $N_2O% emissions for reduction programs.

- **Problem statement:** Both rice and wheat are high emitter in terms of per hectare and total GHG emissions.
- **Objective:** Identify districts with low yields and high n-balance in both rice and wheat as possible district to implement projects as they represent the possible win-wins by improving yields while reducing GHG emissions
- **Logistic:** Identify any spatial cluster of district to minimize logistic complexity.
- **Project Design:** Profile districts to aid in the design of projects to improve yield while reducing GHG emissions

Step 1: imports and setup

In [71]:
```
%reload_ext autoreload
%autoreload 2

get_ipython().run_line_magic('matplotlib', 'inline')
import sys
sys.path.extend(["..\\"])
from Shared import NIBSData2
from Shared.NIBSData2 import *
import duckdb
import geopandas as gpd
import warnings
import skmisc
# Suppress all warnings
warnings.filterwarnings("ignore")

colors = ['#a6cee3','#1f78b4','#b2df8a','#33a02c','#fb9a99','#e31a1c','#fdbf6f','#f
#nibs = NIBSData2.NIBSData()
chart_dir = "./Graphs"
```

Step 2: create in-memory duckdb database of needed parquet table files.

The analysis needs the following tables:

- nitrogen_universe
- data_boundaries
- vwA_apy_adj_crop_area_yield
- n2o_total_direct_n_balance_eagle_2020_summary -state_codes The below code demostrates how to use the helper class `NIBSData2` to load data and get the

In [72]:

```
# Create an in-memory DuckDB connection
con = duckdb.connect(database=":memory:")
con.execute("install spatial")
con.execute("load spatial")
con.execute("SET default_collation = 'nocase';")

# Tables/views listed in the notebook markdown
required_parquets = [
    "nitrogen_universe",
    "data_boundaries",
    "vwA_apy_adj_crop_area_yield",
    "n2o_total_direct_n_balance_eagle_2020_summary",
    "state_codes",
    "national_boundaries",
    "state_boundaries",
    "fertilizer_input",
    "shc_crop_npk_requirements",
    "shc_district_micro_nutrient_summary",
    "shc_parameter_units",
    "state_codes",
    "apy",
]

loaded = []
missing = []

# load non-spatial tables
for name in required_parquets:
    url = nib_urls.get(name)
    if url:
        con.execute(f'''
            CREATE OR REPLACE TABLE "{name}" AS
            SELECT * FROM read_parquet('{url}')
        ''')
        loaded.append(name)
    else:
        missing.append(name)

# load spatial table
# url = nib_urls.get("data_boundaries")
# if url:
#     con.execute(f'''
```

```
#         CREATE OR REPLACE TABLE "data_boundaries" AS
#         SELECT * EXCLUDE geog, (CAST(geog AS string)) AS geog FROM read_parquet('
#     ''')
#     loaded.append("data_boundaries")
# else:
#     missing.append("data_boundaries")

print("Loaded:", loaded)
if missing:
    print("Missing in nib_urls:", missing)

# Optional: verify what is available in the in-memory database
con.sql("SHOW TABLES").df()
```

```
Loaded: ['nitrogen_universe', 'data_boundaries', 'vwA_apy_adj_crop_area_yield', 'n2o
_total_direct_n_balance_eagle_2020_summary', 'state_codes', 'national_boundaries',
'state_boundaries', 'fertilizer_input', 'shc_crop_npk_requirements', 'state_codes',
'apy']
Missing in nib_urls: ['shc_district_micro_nutrient_summary', 'shc_parameter_units']
```

Out[72]:

	name
0	apy
1	data_boundaries
2	fertilizer_input
3	n2o_total_direct_n_balance_eagle_2020_summary
4	national_boundaries
5	nitrogen_universe
6	shc_crop_npk_requirements
7	state_boundaries
8	state_codes
9	vwA_apy_adj_crop_area_yield

Step 3 Build dataset of districts that produce both rice and wheat.

This is a complicated SQL query that use the database created above to find district that product both rice and wheat and return their N_2O emission and spatial bounds data. NOTE the query uses the data_boundaires dataset and not the district_boundaries dataset. This is because the state of Punjab only contains a state-wide result due to the input survey results being incomplete for that state.

In [73]:

```
sql = """
with a as
        (
        select
                r.link_id as rice_id, w.link_id as wheat_id, r.crop, r.apy_crop as
```

```
		, ROW_NUMBER() over(order by w.link_id) as rotation_id
		,ROW_NUMBER() over(partition by r.geog_checksum, r.IRRIGATION_STATU
		,concat(district_name,'(',state_code,')') as label
		,state_code
	from nitrogen_universe as r
		join nitrogen_universe as w on r.geog_checksum = w.geog_checksum
			and r.SIZE_GROUPHA = w.SIZE_GROUPHA
			and r.IRRIGATION_STATUS = w.IRRIGATION_STATUS
			and lower(r.apy_crop) = 'rice'
			and lower(w.apy_crop) = 'wheat'
		join data_boundaries as db on db.geog_checksum = r.geog_checksum
			and db.geog_checksum = w.geog_checksum
		join state_codes as sc on lower(sc.state_name) = lower(db.state_nam
	)
,b as
	(
	select u.link_id
		,a.rotation_id
		, u.geog_checksum
		, label
		,state_code
		, u.apy_crop
		, farm_size
		, irrigated
		, u.adj_crop_area
		, u.adj_fert_area
		, u.mean_yield as mean_yield_t_ha
		, u.sd_yield as sd_yield_t_ha
		,mean_total_n_balance_n_kg_ha
		,sd_total_n_balance_n_kg_ha
	from vwA_apy_adj_crop_area_yield as u
		join n2o_total_direct_n_balance_eagle_2020_summary as n on u.link_i
		join a on a.rice_id = u.link_id
	where dups = 1
	union
	select u.link_id
		,a.rotation_id
		, u.geog_checksum
		, label
		,state_code
		, u.apy_crop
		, farm_size
		, irrigated
		, u.adj_crop_area
		, u.adj_fert_area
		, u.mean_yield as mean_yield_t_ha
		, u.sd_yield as sd_yield_t_ha
		,mean_total_n_balance_n_kg_ha
		,sd_total_n_balance_n_kg_ha
	from vwA_apy_adj_crop_area_yield as u
		join n2o_total_direct_n_balance_eagle_2020_summary as n on u.link_i
		join a on a.wheat_id = u.link_id
	where dups = 1
	)
,c as
	(
```

```
        select *
                ,count(*) over(partition by rotation_id) as rcnt
        from b
        )
select c.*
        ,cast(geog as string) as geog
        ,label_x
        ,label_y
from c
        join data_boundaries as d on d.geog_checksum = c.geog_checksum
where rcnt = 2;
"""

# table = ['nitrogen_universe', 'data_boundaries', 'vwA_apy_adj_crop_area_yield', '

# for t in table:
#       print(f"Sample from {t}:")
#       print(con.execute(f"SELECT * FROM {t} LIMIT 5").fetch_df())

# sql = "select geog from data_boundaries limit 5"

df = con.execute(sql).fetch_df()
rice_wheat_gdf = gpd.GeoDataFrame(df,geometry= gpd.GeoSeries.from_wkt(df['geog']),c
rice_wheat_gdf.head(3)
```

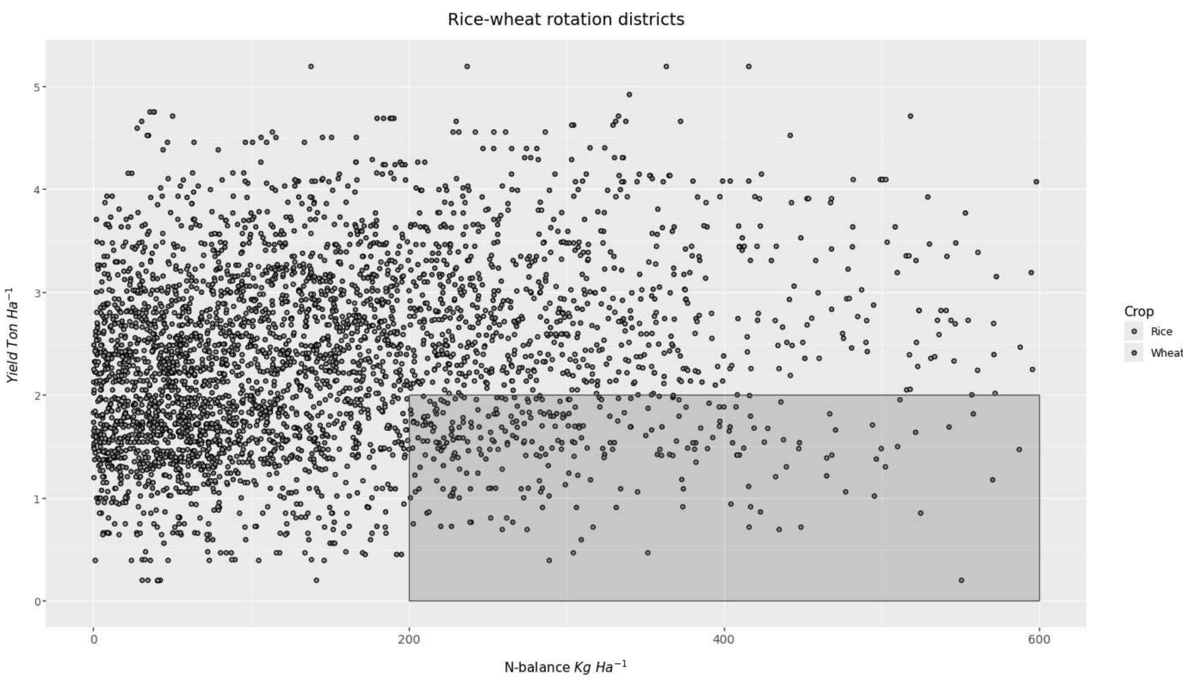
Out[73]:

	link_id	rotation_id	geog_checksum	label	state_code	apy_crop
0	-9004847403858430320	80	1981673543	Jyotiba Phule Nagar(UP)	UP	Wheat
1	-2556190698396902204	80	1981673543	Jyotiba Phule Nagar(UP)	UP	Rice
2	-8992185940145162137	86	-685168729	Lalitpur(UP)	UP	Wheat

Step 4: District yield vs n-balance graph

This graph, graphs the district yield on the y-axis and the n-balance on the x-axis for both rice and wheat. The bottom quadrant of the graph respresents districts with high n-balance (over application of N) and low yields. These districts can be targeted for triple-wins, lowing N application and N_2O emissions, improving yeilds and farmer livehoods hilighted in the red box on the graph.

In [74]:
```
p = (ggplot(rice_wheat_gdf)
            + geom_point(aes(x='mean_total_n_balance_n_kg_ha', y='mean_yield_t_ha',
            + scale_fill_discrete(name='Crop')
             + labs(title='Rice-wheat rotation districts', x="N-balance $Kg\ Ha^{-1
                 + annotate(
                "rect",
                xmin=200, xmax=600,
                ymin=0, ymax=2,
                fill="red",
                alpha=0.2,      # keep it translucent so data underneath stays visi
                color="red",    # red border
            )
            + scale_x_continuous(limits=(0, 600))
            #+ guides(fill=None)
             + theme(figure_size=(14,8), axis_text_x=element_text(angle=0, va="top"
             plot_title=element_text(ha='center', size=14))
            )

p.save(filename=f"rice_wheat_n_balance_target_full.png", path=chart_dir, height=12,
p.show()
```

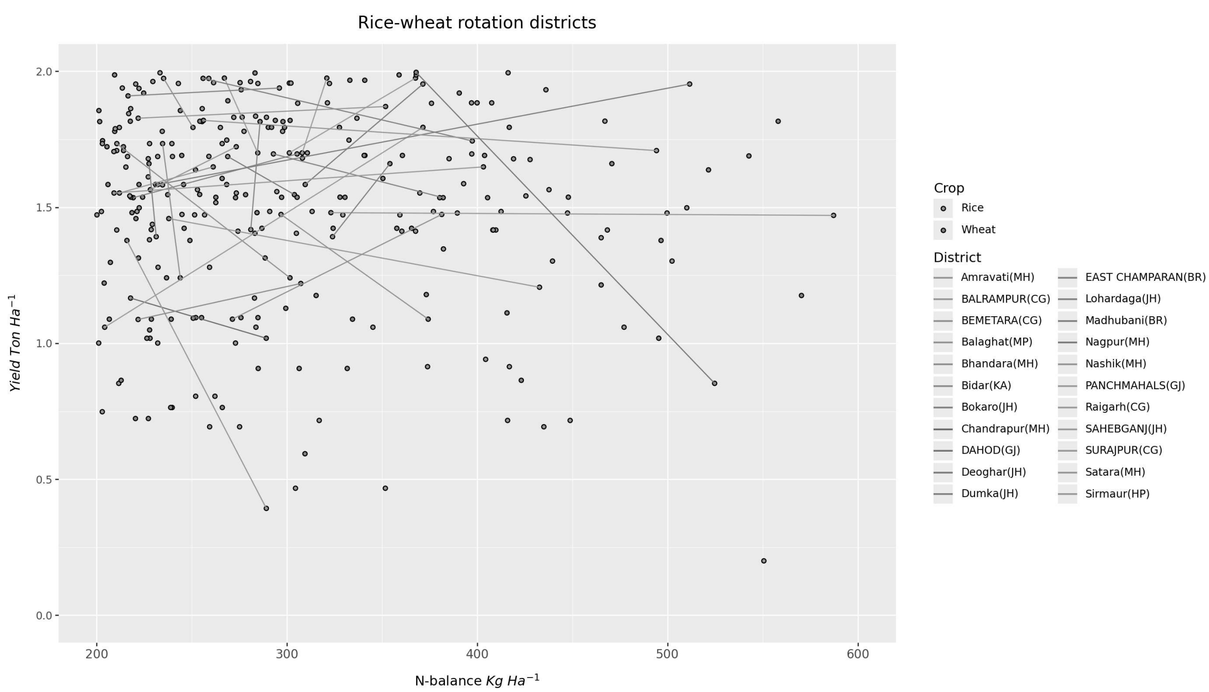


Step 5: Identity target districts.

Using what we have learned from the graph above we will now find district where both rice and wheat fall within the target zone identified above.

In [75]:
```
#filter the rice-wheat rotation districts that are in the target zone (N-balance >
filter_gdf = rice_wheat_gdf[(rice_wheat_gdf['mean_total_n_balance_n_kg_ha'] > 200)

#create the dataset the links the rice and wheat for a districts together by rotati
grouped_df = filter_gdf.groupby('rotation_id').size().reset_index(name='counts')
line_rid = grouped_df[grouped_df['counts'] > 1]['rotation_id']
```

```
line_df = filter_gdf[filter_gdf['rotation_id'].isin(line_rid)]
line_df['mean_total_n_balance_n_kg_kg_yield'] =line_df['mean_total_n_balance_n_kg_h
line_cnt_gdf = line_df.groupby('label').size().reset_index(name='counts')
line_cnt_gdf

#graph the rice-wheat rotation districts that are in the target zone (N-balance > 2
p = (ggplot(filter_gdf)
            # + geom_violin(filter_df,aes(x='dataset', y='kg_inorganic_n_ha'),draw_
            + geom_point(aes(x='mean_total_n_balance_n_kg_ha', y='mean_yield_t_ha',
        #     + geom_errorbar(aes(x='label_simple', ymin='mean_co2e_Mt - sd_co2e_Mt
        #   + geom_text(aes(x='label_simple', y='mean_co2e_Mt', label='mean_co2e_Mt
        #         va='bottom', ha='center', size=8, format_string='{:.2f}')

            # + geom_jitter(filter_df, aes(x='dataset', y='kg_inorganic_n_ha'), col
             + labs(title='Rice-wheat rotation districts', x="N-balance $Kg\ Ha^{-1
            # + scale_y_log10()
            + scale_fill_discrete(name='Crop')
            + scale_y_continuous(limits=(0, 2))
            + scale_x_continuous(limits=(200, 600))
            + geom_line(line_df, aes(x='mean_total_n_balance_n_kg_ha', y='mean_yiel

            + scale_color_discrete(name='District')
            #  +scale_y_continuous(limits=(0, 375))
            #+ guides(fill=None)
             + theme(figure_size=(14,8),axis_text_x=element_text(angle=0, va="top",
             plot_title=element_text(ha='center', size=14))
            # + facet_wrap('~gwp_label')
            )
p.save(filename=f"rice_wheat_n_balance_target_zoom.png", path=chart_dir, units='cm'
p
```

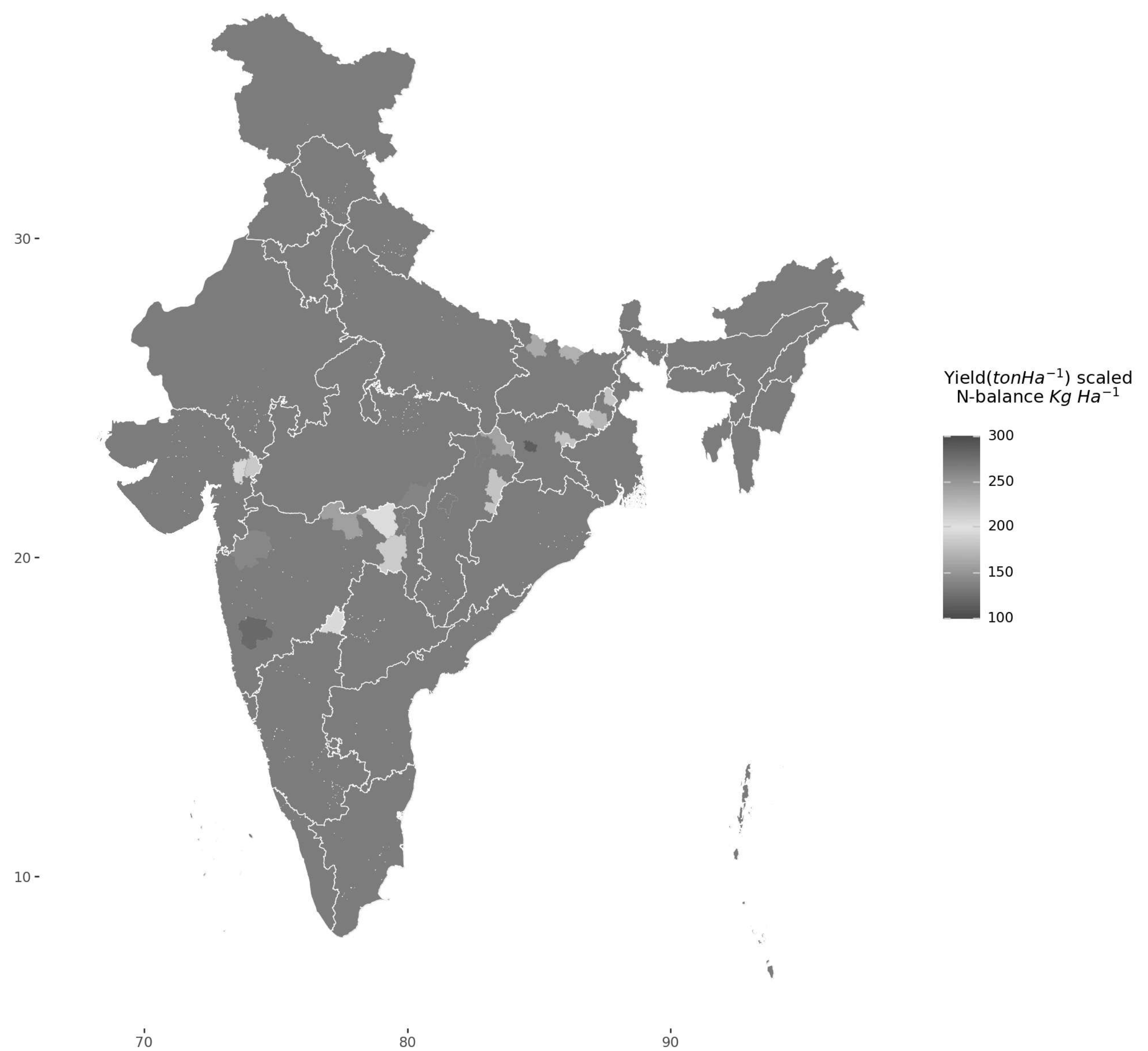


Step 6: Identify spatial clustering

Now we would like to look for any spatial clustering of districts that would be ideal for developing a project in to reduce emissions of N_2O.

In [81]:
```
#load the India national and state boundaries
india_df = con.execute("SELECT * EXCLUDE geog, (CAST(geog AS string)) AS geog FROM
india = gpd.GeoDataFrame(india_df, geometry=gpd.GeoSeries.from_wkt(india_df['geog']
state_df = con.execute("SELECT * EXCLUDE geog, (CAST(geog AS string)) AS geog FROM
india_states = gpd.GeoDataFrame(state_df, geometry=gpd.GeoSeries.from_wkt(state_df[

#create map graph
units = "Yield($ton Ha^{-1}$) scaled\nN-balance $Kg\ Ha^{-1}$\n"
low_color = "green"
mid_color= "yellow"
high_color = "red"
g = (ggplot()
     + geom_map(india, fill='grey', color=None, show_legend=False)
     #+ scale_fill_brewer(type='div', palette=8)
     + scale_x_continuous(limits=(67.5,97.5))
     #+ scale_y_continuous(limits=(7.5,37.5))
     + coord_cartesian()
     + geom_map(line_df, aes(fill="mean_total_n_balance_n_kg_kg_yield"), color=None
     + geom_map(india_states, color="white", fill=None, size=.25, show_legend=False
     + labs(title='Rice-wheat rotation districts')
     + scale_fill_gradient2(low=low_color, mid=mid_color, midpoint=200, high=high_c
     + theme(
        figure_size=(10, 10),
        rect=element_rect(fill=(0, 0, 0, 0), color=(0, 0, 0, 0)),
        strip_background=element_rect(fill='white', color='white'),
        plot_title=element_text(ha='center', size=14)
     )
)

#g.save(filename="map_rice_wheat_targets.png", path=chart_dir,  units='cm', dpi=300
g
```

Rice-wheat rotation districts



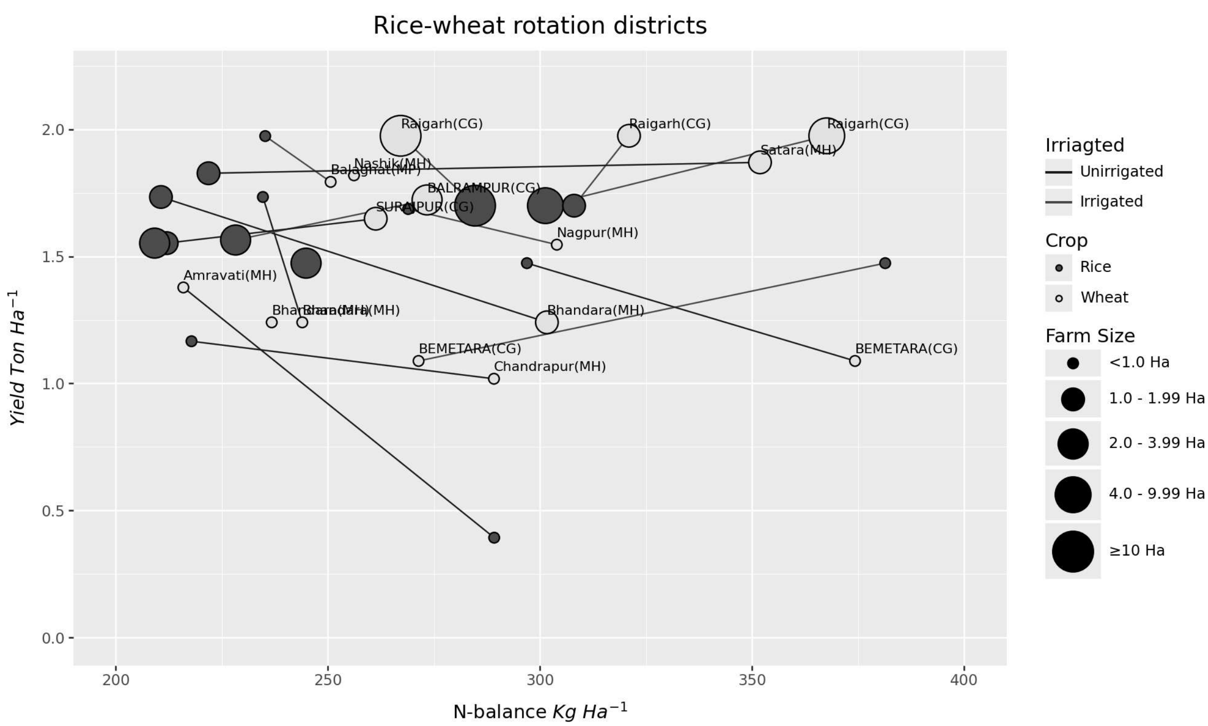

```
In [ ]:
```

Step 7: Profile target districts

Once the target district for a project are identified we can now profile them to better understand their production attributes. We will start with profiling their irrigation status and farm size for those with low yields and high N_2O.

```
In [10]: #create a dataset of spatially clustered rice-wheat rotation districts in the targe
target_df = line_df[(line_df['state_code'] == 'MH') | (line_df['state_code'] == 'MP
target_df.loc[target_df['apy_crop'] == 'Rice', 'label'] = ''
target_df
target_df['farm_size'] = pd.Categorical(target_df['farm_size']
                                    , categories=[ 'MARGINAL (BELOW 1.0)', 'SMALL (
target_df['farm_size'] = target_df['farm_size'].cat.rename_categories({'MARGINAL (B
                                                                       'SMALL (1.0
                                                                       'SEMI-MEDIU
```

```
                                                                   'MEDIUM (4.
                                                                   'LARGE (10
target_df['irrigated'] = pd.Categorical(target_df['irrigated']
                                    , categories=['UI','I'], ordered=True)
target_df['irrigated'] = target_df['irrigated'].cat.rename_categories({'UI': 'Unirr
target_df


#graph the rice-wheat rotation districts use color to represent crop (Rice or Wheat
g = (ggplot(target_df,aes(x='mean_total_n_balance_n_kg_ha', y='mean_yield_t_ha'))
    + geom_line(aes(group='rotation_id', color='irrigated'))
    + geom_point(aes(fill='apy_crop', size='farm_size'))
    + geom_text(aes( label='label', adjust_text=True), va='bottom', ha='left', size
    + labs(title='Rice-wheat rotation districts', x="N-balance $Kg\ Ha^{-1}$", y="$

     + scale_color_manual(name='Irriagted',values={'Unirrigated': 'blue', 'Irrigate
     + scale_fill_manual(name='Crop',values={'Rice': 'red', 'Wheat': 'yellow'})
     + scale_size_discrete(name='Farm Size', range=[3,13])
     + scale_x_continuous(limits=(200, 400))
     + scale_y_continuous(limits=(0, 2.2))
     + theme(
         figure_size=(10, 6),
        # rect=element_rect(fill=(0, 0, 0, 0), color=(0, 0, 0, 0)),
         #strip_background=element_rect(fill='white', color='white'),
         plot_title=element_text(ha='center', size=14)
     )
)
g.save(filename="rice_wheat_targets_detail.png", path=chart_dir, units='cm', dpi=30
g
```



Step 8: Profile district yield and crop area.

Profile the target districts to see their yield for rice and wheat overtime compared to the all India average. Explore if there is enough crop area to implement a project and achieve the required scale.

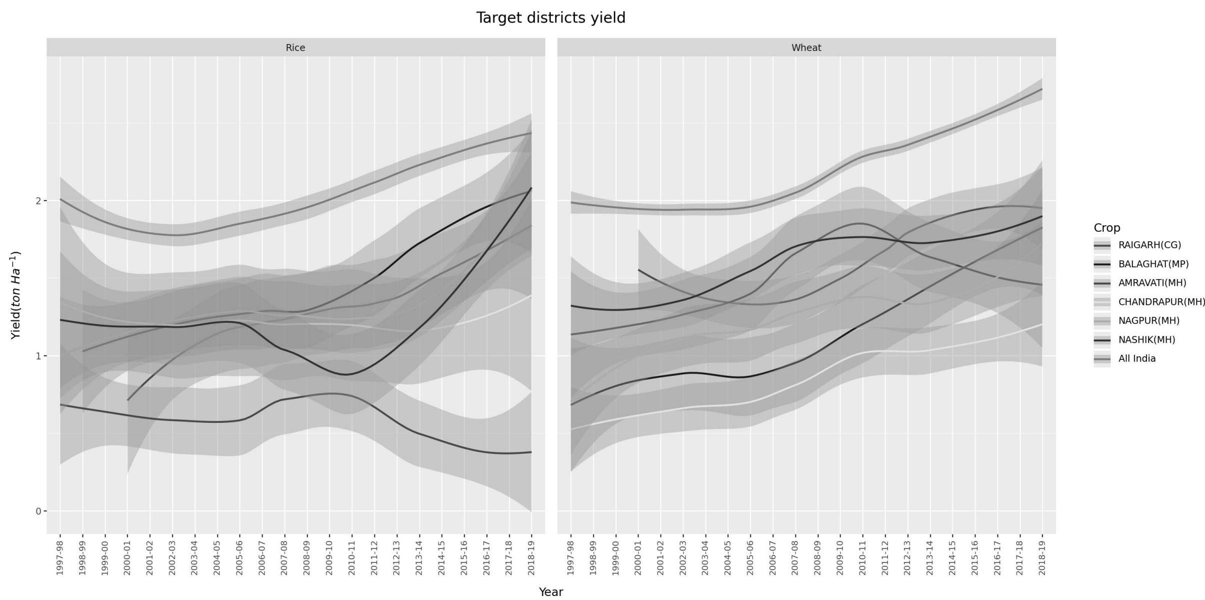
In [77]:
```
apy_sql = """with a as
        (
        select concat(district_name,'(',state_code,')') as district
                ,apy_crop, year
                ,area, production, yield
                ,season
                ,row_number() over(partition by apy.state_name,district_name,apy_cr
        from apy
                join state_codes sc on sc.state_name = apy.state_name
        where geog_checksum in(357371350,-2066345052,931355363,1626958676,207678571
                and apy_crop in('rice','wheat')
        )
,b as (
        select 'All India' as district
                ,apy_crop, year
                ,area, production, yield
                ,season
                ,row_number() over(partition by apy.state_name,district_name,apy_cr
        from apy
                join state_codes sc on sc.state_name = apy.state_name
        --where geog_checksum not in(357371350,-2066345052,931355363,1626958676,207
                and apy_crop in('rice','wheat')
        and year <> '2019-20'
    )
select district, apy_crop, year, area, production, yield
from a
where rno = 1
union
select district, apy_crop, year, area, production, yield
from b
where rno = 1
"""
apy_df = con.execute(apy_sql).fetch_df()

apy_df['district'] = pd.Categorical(apy_df['district'], categories=['RAIGARH(CG)',
apy_df.head()
```

Out[77]:

	district	apy_crop	year	area	production	yield
0	RAIGARH(CG)	Wheat	2001-02	1339.0	1892.0	1.412995
1	BALAGHAT(MP)	Rice	2006-07	248652.0	299342.0	1.203859
2	BALAGHAT(MP)	Rice	2018-19	300301.0	603446.0	2.009470
3	BALAGHAT(MP)	Rice	2004-05	224800.0	246491.0	1.096490
4	CHANDRAPUR(MH)	Wheat	2014-15	24200.0	17800.0	0.735537

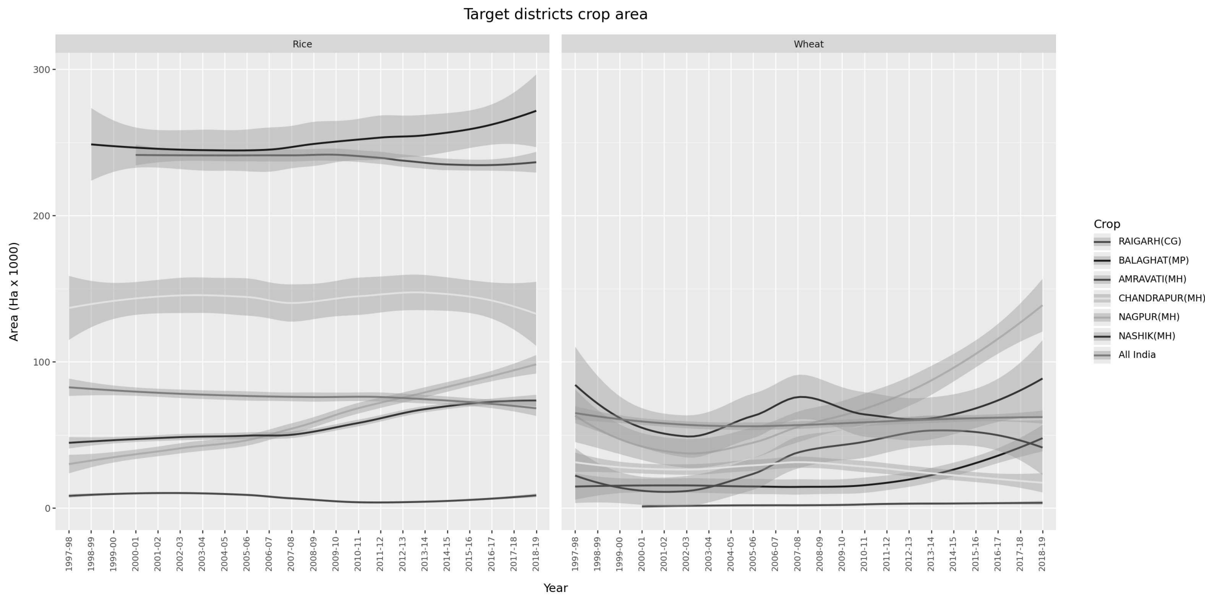
Create graph of target district yields over time against the all India average. The graph shows the target districts are below all India average with different trends.

```
In [78]: g = (ggplot(apy_df)
             + geom_smooth(aes(x='year', y='yield', color='district', group='district'), met
             + labs(title='Target districts yield', x='Year', y='Yield($ton\ Ha^{-1}$)')
             + scale_color_manual(name='Crop', values={'RAIGARH(CG)': 'red', 'BALAGHAT(MP)':
             + theme(
                 figure_size=(16, 8),
                 axis_text_x=element_text(angle=90, va="top", ha="center", size=8),
                 plot_title=element_text(ha='center', size=14)
             )
             + facet_grid('.~apy_crop')
         )
         g.save(filename="target_district_yield.png", path=chart_dir, units='cm', dpi=300)
         g
```



Create a graphs of cropping area overtime to show the scale possible projects would achieve within this

```
In [79]: g = (ggplot(apy_df)
             + geom_smooth(aes(x='year', y='area/1000', color='district', group='district'),
             + labs(title='Target districts crop area', x='Year', y='Area (Ha x 1000)')
             + scale_color_manual(name='Crop', values={'RAIGARH(CG)': 'red', 'BALAGHAT(MP)':
             + theme(
                 figure_size=(16, 8),
                 axis_text_x=element_text(angle=90, va="top", ha="center", size=8),
                 plot_title=element_text(ha='center', size=14)
             )
             + facet_grid('.~apy_crop')
         )
         g.save(filename="target_district_area.png", path=chart_dir, units='cm', dpi=300)
         g
```


Target districts crop area
Rice
Wheat
Area (Ha x 1000)
300
200
100
0
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
2015-16
2016-17
2017-18
2018-19
Year
Crop
RAIGARH(CG)
BALAGHAT(MP)
AMRAVATI(MH)
CHANDRAPUR(MH)
NAGPUR(MH)
NASHIK(MH)
All India